Download and Install Python on Windows, Linux and macOS

This lab walks you through installing Python 3 on your own computer, confirming it works, and creating a first virtual environment. Complete only the part for your operating system, then everyone finishes with Part 4.

TIME	LEVEL	YOU NEED
20–30 min	Beginner	Admin rights + internet

Objectives

By the end of this lab you will be able to:

1. Download the official Python 3 installer or package for your system.
2. Install Python so it can be run from a terminal.
3. Verify the Python and pip versions.
4. Create and activate a virtual environment and install a package into it.

Before you begin

- Always download Python from python.org/downloads or your system's package manager — never from third-party mirror sites.
- Install the latest stable **Python 3** release. Python 2 is end-of-life and must not be used.
- Commands in this guide are shown in boxes. Type them exactly; lines starting with # are comments.

SYSTEM	MINIMUM	TERMINAL TO USE
Windows	Windows 10 or 11, 64-bit	PowerShell or Windows Terminal
Linux	Any current distribution	Terminal (bash/zsh)
macOS	A supported macOS release (Intel or Apple silicon)	Terminal.app

PART 1

Windows

Step 1.1 — Download the installer

1. Open python.org/downloads/windows.
2. Under **Stable Releases**, choose the newest Python 3 release.
3. Download **Windows installer (64-bit)**. Use *ARM64* only if your PC has an ARM processor.

Step 1.2 — Run the installer

1. Double-click the downloaded `python-3.x.x-amd64.exe`.
2. On the first screen, tick **Add python.exe to PATH**.
3. Click **Install Now** and approve the administrator prompt.
4. When setup finishes, click **Disable path length limit** if offered, then **Close**.

> IMPORTANT
>
> Forgetting **Add python.exe to PATH** is the most common setup error. If you missed it, re-run the installer, choose *Modify*, and enable *Add Python to environment variables*.

Step 1.3 — Verify

Open a **new** PowerShell window and run:

POWERSHELL

```
python --version
py --version
python -m pip --version
```

Each command should print a version number such as `Python 3.x.y`. If `python` opens the Microsoft Store instead, see Troubleshooting.

Alternative — install with winget

On Windows 10/11 with the App Installer, you can install from the command line instead:

POWERSHELL

```
# list available Python packages, then install the newest one shown
winget search Python.Python
winget install Python.Python.3.14
```

PART 2

Linux

Most distributions ship with Python 3 already. Use the system package manager to install or update it — do not remove the system Python, as the OS depends on it.

Step 2.1 — Check what is installed

TERMINAL

```
python3 --version
```

Step 2.2 — Install Python, pip and venv

Run the block that matches your distribution:

TERMINAL

```
# Ubuntu / Debian / Linux Mint
sudo apt update
sudo apt install python3 python3-pip python3-venv

# Fedora / RHEL / Rocky / AlmaLinux
sudo dnf install python3 python3-pip

# Arch / Manjaro
sudo pacman -S python python-pip

# openSUSE
sudo zypper install python3 python3-pip
```

Step 2.3 — Verify

TERMINAL

```
python3 --version
python3 -m pip --version
```

Alternative — a newer version on Ubuntu

If your Ubuntu release ships an older Python than your course requires, the deadsnakes PPA provides newer builds alongside the system one:

TERMINAL

```
sudo add-apt-repository ppa:deadsnakes/ppa
sudo apt update
sudo apt install python3.14 python3.14-venv
python3.14 --version
```

PART 3 macOS

macOS may include a `python3` stub from Apple's Command Line Tools, but it is often outdated. Install the official release instead.

Step 3.1 — Download the installer

1. Open python.org/downloads/macos.
2. Download the **macOS 64-bit universal2 installer** (`.pkg`) for the newest Python 3 release. It runs natively on both Intel and Apple silicon Macs.

Step 3.2 — Run the installer

1. Open the `.pkg` file and click **Continue** through each screen.
2. Accept the licence, click **Install**, and enter your Mac password.
3. When the Finder window for *Applications › Python 3.x* opens, double-click **Install Certificates.command**.

IMPORTANT

Skipping **Install Certificates.command** causes *SSL: CERTIFICATE_VERIFY_FAILED* errors when Python downloads from the internet.

Step 3.3 — Verify

Open a **new** Terminal window. On macOS, always type `python3`, not `python`.

TERMINAL

```
python3 --version
which python3        # should show /Library/Frameworks/... or /usr/local/bin/...
python3 -m pip --version
```

Alternative — install with Homebrew

If you already use Homebrew, it can install and update Python for you. Use one method only — not both.

TERMINAL

```
brew update
brew install python
python3 --version
```

PART 4 Test your installation (all systems)

In the commands below, Windows users type `python`; Linux and macOS users type `python3`.

Step 4.1 — Run a first program

Create a file named `hello.py` containing:

HELLO.PY

```
import sys
print("Hello from Python", sys.version.split()[0])
```

Then run it from the same folder:

TERMINAL

```
python3 hello.py
```

Step 4.2 — Create a virtual environment

A virtual environment keeps each project's packages separate from the system Python.

TERMINAL

```
mkdir python-lab
cd python-lab
python3 -m venv .venv

# activate — Linux / macOS
source .venv/bin/activate

# activate — Windows PowerShell
.venv\Scripts\Activate.ps1
```

Your prompt now begins with `(.venv)`.

Step 4.3 — Install a package

TERMINAL (INSIDE .VENV)

```
python -m pip install --upgrade pip
python -m pip install requests
python -c "import requests; print(requests.__version__)"
deactivate
```

REFERENCE

Troubleshooting

SYMPTOM	FIX
Windows: *'python' is not recognized*	PATH was not set. Re-run the installer › Modify › enable *Add Python to environment variables*, then open a new terminal. Or use `py`.
Windows: `python` opens the Microsoft Store	Settings › Apps › Advanced app settings › App execution aliases › turn off the two *python* aliases.
Windows: *Activate.ps1 cannot be loaded*	Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, then activate again.
Linux: *externally-managed-environment* error from pip	Expected on modern distributions. Install packages inside a virtual environment (Step 4.2), not system-wide.
Linux: *ensurepip is not available*	Install the venv package, e.g. `sudo apt install python3-venv`.
macOS: *CERTIFICATE_VERIFY_FAILED*	Run *Applications › Python 3.x › Install Certificates.command*.
macOS: old version still reported	Open a new Terminal, check `which -a python3`, and make sure the new path comes first.

SIGN-OFF

Lab checklist

- ☐ Python 3 is installed and `--version` prints a 3.x number.
- ☐ pip is available via `-m pip --version`.
- ☐ `hello.py` runs and prints its version.
- ☐ A `.venv` was created, activated, and `requests` installed into it.

Name	Operating system / Python version	Date